$\neg P$

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Date :

Q1 Define Partition of a set with a suitable example.

→ A partition of a set divides set into non-overlapping non-empty subsets whose union equals the original sets. Essentially it's a way to break down a set into smaller, complete parts.

Example:

Consider the set $S = \{1, 2, 3, 4, 5, 6\}$. One possible partition of S could be

$A = \{1, 2, 3\}$

$B = \{4, 5\}$

$C = \{6\}$

Q2) Simplify using Laws of logic

$(\neg P \wedge Q) \vee (\neg P \wedge \sim Q) \Leftrightarrow \neg P \wedge (Q \vee \sim Q)$

$\Rightarrow (\neg P \wedge Q) \vee (\neg P \wedge \sim Q) \Leftrightarrow \neg P \wedge (Q \vee \sim Q)$

Using Distributive Law

$\neg P \wedge (Q \vee \neg Q)$

Using Complement Law ... $(A \vee \neg A) \Leftrightarrow T$

$(Q \vee \neg Q) \Leftrightarrow$

$\therefore \neg P \wedge (Q \vee \neg Q) \Leftrightarrow \neg P \wedge T$

$\neg P \wedge (Q \vee \neg Q) \Leftrightarrow \neg P$ ... (Identity law $A \wedge T = A$)

Q3. Define Power set with a suitable example.

→ The power set of a given set is the set of all possible sub-sets of that sets, including the empty set and the set itself. In simpler terms, it's set containing every combination of elements from the original set, where order doesn't matter.

Example :

$A = \{1, 2\}$

The power set of A, denoted as P(A)

$P(A) = \{ \{\}, \{1\}, \{2\}, \{1, 2\} \}$

~~Define Equivalence Relation. Let R is defined by (a,b) R (c,d) if and only if a+d = b+c. Prove that R is an equi~~

Define and explain Quantifier with an example.

→ Quantifier are words or symbols that specify the quantity or extent of something. They indicate how many elements of a set satisfy particular condition.

- Universal Quantifier ($\forall$):

This quantifier means "for all" or "every". It asserts that a property holds true for every element in a set or domain.

e.g. ~~∀(x~~ $\forall x$ (x is mammal $\longrightarrow$ x has fur).

Existential Quantifier ($\exists$):
This quantifier means "there exists" or "at least one". It asserts that there is at least one element in a set or domain that satisfies a particular property
ex- $\exists x$ (x is a prime number $\wedge$ $x > 10$)

Q9) Same as question no. 8.

Q10) Symbolize using quantifiers:

1. All students took mathematics.
→ $\forall x\,(S(x) \longrightarrow M(x))$
where ($S(x)$) represent x is a student.
$M(x)$ = student (x) took mathematics

2. There is a girl student whom is a sportperson
→ $\exists x\,(G(x) \wedge S(x))$
where $G(x)$ = "x" is a girl student
$S(x)$ = "x" is a sportperson

3. Some students are intelligent, but not hardworking
→ $\exists(x)\,(S(x) \wedge I(x) \wedge \neg H(x))$
where $S(x)$ is x student
$I(x)$ = x is ~~intelligent~~ intelligent student
$H(x)$ = x is hardworking.

Construct a truth table and verify whether $(P \rightarrow Q) \longleftrightarrow (\neg Q \rightarrow \neg P)$ is a tautology

P	Q	$P \rightarrow Q$ (I)	$\neg Q$	$\neg P$	$\neg Q \rightarrow \neg P$ (II)	$I \leftrightarrow II$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Since final column of the truth table $(P \rightarrow Q) \longleftrightarrow (\neg Q \rightarrow \neg P)$ is 'True'. Therefore expression is tautology.

6)

Q15. Construct a truth table for
$(P \rightarrow Q) \wedge (\sim Q \vee R)$

P	Q	R	$P \rightarrow Q$ (I)	$\sim Q$	$\sim Q \vee R$ (II)	(I) ∧ (II)
T	T	T	T	F	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	F
T	F	F	F	T	T	F
F	T	T	T	F	T	T
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	F

Q16) Define power set with suitable example.
→ Same as question no. 3.

Q17) Construct truth table and check whether
$(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$ is tautology.

P	Q	$P \rightarrow Q$ (I)	$\neg Q$	$\neg P$	$\neg Q \rightarrow \neg P$ (II)	I ↔ II
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Since final column of truth table is true
∴ Therefore this expression is tautology.

Date :

Q21 Prove by mathematical induction.

$2+5+8+\ldots+(3n-1) = \frac{n(3n+1)}{2}$

→ ~~n=~~ Let $n=1$

$LHS = (3n-1)$

$(3\times1-1) = 2$

$RHS = \frac{n(3n+1)}{2}$

$= \frac{1(3\times1+1)}{2}$

$= 4/2$

$= 2$

$\therefore LHS = RHS.$

Consider $n=k$

~~LHS~~

$2+5+8+\ldots+(3k-1) = \frac{k(3k+1)}{2}$

Now, $n=k+1$

$2+5+8+\ldots+(3(k+1)-1) = \frac{(k+1)(3(k+1)+1)}{2}$

LHS is.

$2+5+8+(3k-1)+3(k+1)-1)$

The sum of first k terms is $\frac{k(3k+1)}{2}$

$\therefore \frac{k(3k+1)}{2} + (3(k+1)-1)$

$$= \frac{k(3k+1)}{2} + (3k+3-1)$$

$$= \frac{k^2+k+2(3k+2)}{2}$$

$$= \frac{k^2+k+6k+4}{2}$$

$$LHS = \frac{k^2+7k^2+4}{2}$$

Now, simplify RHS of $n=k+1$

$$= \frac{(k+1)(3(k+1)+1)}{2}$$

$$= \frac{(k+1)(3k+3+1)}{2}$$

$$= \frac{(k+1)(3k+4)}{2}$$

$$= \frac{3k^2+4k+3k+4}{2}$$

$$RHS = \frac{3k^2+7k+4}{2}$$

$\therefore$ LHS = RHS.

22. Prove by Mathematical induction.

$1!+2.2!+3.3!+\ldots+n.n! = (n+1)!-1$

→ Let $n=1$

$LHS = n.n! = 1.1! = 1$

$RHS = (n+1)!-1 = (1+1)!-1$

$= 2!-1 =$

$RHS = 1$

$\therefore LHS = RHS.$

Consider $n=k$

$1*1!+2.2!+3.3!+k.k! = (k+1)!-1$

Now, Let $n=k+1$

$1!+2.2!+3.3!+k.k!+(k+1).(k+1)! = (k+2)!-1 \ldots (i)$

$LHS = 1!+2.2!+3.3!+k*k!+(k+1)*(k+1)!$

Sum of first k terms is $(k+1)!-1$

$\therefore (k+1)!-1+(k+1).(k+1)!$

$= (k+1)!\,(1+(k+1+1)-1$

$= (k+1)!\,(k+2)-1$

$(k+1)!\,(k+2)$ is factorial of $(k+2)!$

$= (k+2)!-1 \quad \ldots$ LHS = RHS ..(from eqn 1)

This is equivalent to RHS of eqn for $n=k+1$